SAFETY BAY
SENIOR HIGH SCHOOL
imagine believe achieve

Sem 2 Examination 2015

Question/answer booklet

YEAR 10A EXTENSION MATHEMATICS

Name: Mark Scheme

Teacher: Roy

Section One: Calculator-free

Time allowed for this section

Section One

Reading time before commencing work: 2 minutes
Working time for paper: 28 minutes

Material required/recommended for this paper

To be provided by the supervisor

This question/answer Booklet
Formula sheet

To be provided by the student

Standard items: pens, pencils, pencil sharpener, eraser, correction fluid/tape, ruler, highlighter

Standard items: nil

Important note to students

This test comprises of **TWO sections**. The **first section** is **calculator free** where no calculators of any kind are to be used. The **second section** is **calculator assumed** where a scientific calculator may be used. All questions must be answered in both sections. **Answers should be rounded appropriately**. All working should be shown in the space provided. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

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MARK ALLOCATION AND RECORDS:

Section	Question	Marks	Awarded	
ONE	1	3		
	2	5		
	3	4		
	4	3		
	5	5		
	6	2		
	7	2		
	8	3		
	ONE	**27**		%
	TWO	**33**		%
	TOTAL	**60**		%

Question 1 (3 marks: 1, 2)

Simplify the following

a) $\sqrt{500}$

$\sqrt{5 \times 100}$

$= 10\sqrt{5}$ ✓

f) $\log_7 4 + \log_7 12 - \log_7 3$

$\frac{4 \times \cancel{12}4}{\cancel{3}} = 16$ ✓ ✓

Question 2 (5 marks: 1, 2, 2)

Simplify the following, leaving your answers with positive indices, or evaluate.

a) $2^8 \times 2^{11}$

2^{19} ✓

b) $\frac{\overset{4}{\cancel{64}}\cancel{a^4}\cancel{y^2}z^2}{\cancel{16}a^5y^3} = \frac{4z^2}{ay}$ ✓ ✓

f) $5^2 + (x^5 + 5)^0$

$25 + 1 = 26.$ ✓ ✓

Question 3 (4 marks: 2, 2)

Expand and where possible, simplify the following

a) $p(p - 3)$

$p^2 - 3p$

✓ ✓

c) $(q + 5)^2$

$(q+5)(q+5) = q^2 + 5q + 5q + 25$ ✓

$= q^2 + 10q + 25$ ✓

Question 4 (3 marks: 1, 2)

Rationalise the denominator in the following

a) $\frac{4}{\cancel{\sqrt{7}}} \times \frac{\cancel{\sqrt{7}}}{\sqrt{7}}$

$= \frac{4\sqrt{7}}{7}$

Check... $= \frac{4}{\sqrt{7}}$ ✓

b) $\frac{\sqrt{6}}{\sqrt{2}} \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}\sqrt{2}}{2}$ ✓

$= \frac{\sqrt{3 \times 2}\sqrt{2}}{2}$

or $\frac{\sqrt{3}\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}}$ ✓

$= \sqrt{3}$ ✓

OR

$= \frac{\sqrt{3} \times \cancel{\sqrt{2}} \times \cancel{\sqrt{2}}}{\cancel{2}}$

$= \sqrt{3}$ ✓

15

Question 5 (4 marks: 2, 3)

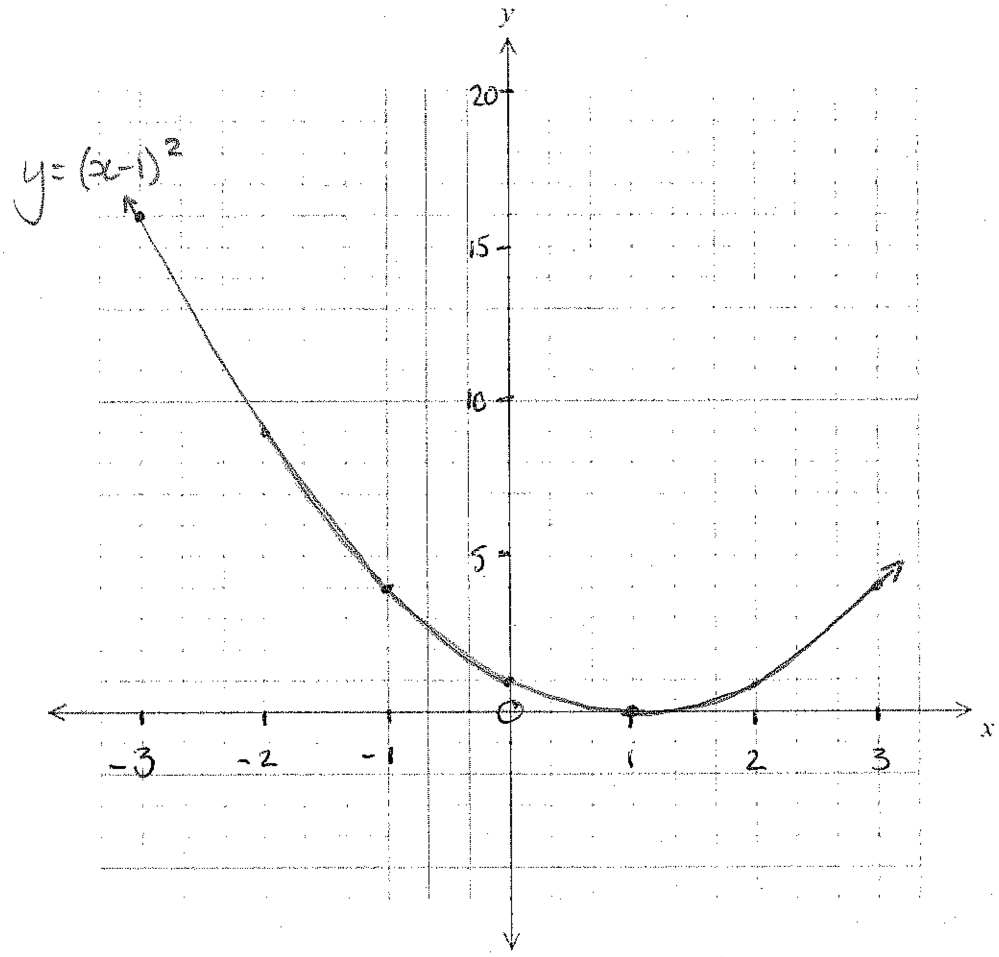
For the equation $y = (x-1)^2$

a) Complete the table of values for $-3 \le x \le 3$

x	−3	−2	−1	0	1	2	3
y	16	9	4	1	0	1	4

√√

b) Plot the function on the axes below; (use 3 squares for one unit on the x axis)



√ curve not straight lines

√ spacing on x axis

√ intercept and t.p.

Question 6 (2 marks: 1, 1)
Evaluate the following

a) $\log_2 32 = 5$

$2^x = 32$

$x = 5$ ✓

b) $\log_5 125 = 3$

$5^x = 125$

$x = 3$ ✓

Question 7 (2 marks)
Solve this set of simultaneous equations

$$y = 3x - 7$$
$$y = 3 - 2x$$

$3x - 7 = 3 - 2x$

$5x - 7 = 3$

$5x = 10$

$x = 2$ ✓

$y = -1$ ✓

$(2, -1)$

Question 8 (3 marks: 1, 2)
Solve the following

a) $2^x = 64$

$2^6 = 64$

$\therefore x = 6$ ✓

b) $3^{x-1} = 81$

$3^4 = 81$

$x - 1 = 4$ ✓

$\underline{\underline{x = 5}}$ ✓

End of Section One

Spare page

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Sem 2 Examination 2015
Question/answer booklet

YEAR 10A
EXTENSION
MATHEMATICS

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Teacher: Roy

Section Two:
Calculator-assumed

Time allowed for this section
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This question/answer Booklet
Formula sheet

To be provided by the student
Standard items: pens, pencils, pencil sharpener, eraser, correction fluid/tape, ruler, highlighter

Special items: *drawing instruments, templates, notes on two unfolded sheets of A4 paper, and scientific calculator*

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	10	2		
	11	3		
	12	2		
	13	2		
	14	7		
	15	2		
	16	3		
	17	5		
	18	3		
	ONE	**27**		%
	TWO	**33**		%

TOTAL	**60**		%

Question 9 (3 marks: 1, 2)
Factorise the following

a) $12x + 36$

$12(x+3)$ ✓

b) $x^2 + 9x + 14$

$(x+2)(x+7)$ ✓ ✓

Question 10 (2 marks: 1, 1)
Calculate to 2 decimal places the following

a) $\log_5 29$

$= \frac{\log 29}{\log 5} = 2.092$

$\approx \underline{\underline{2.09}}$

b) $\log_2 306$

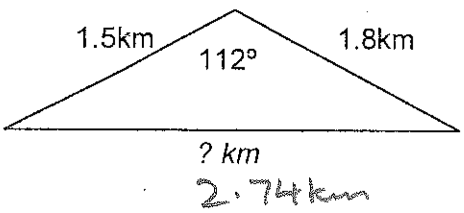
$\frac{\log 306}{\log 2} = 8.2573$

$\approx \underline{\underline{8.26}}$

Question 11 (3 marks)

A speed boat race is run on a triangular course on a harbor. Two sides of the course measure 1.5 km and 1.8 km, and create an angle of 112°.

What is the length of the longest side in the triangle that makes up this race course?



$a^2 = b^2 + c^2 - 2bc\cos A$

$= (1.5^2 + 1.8^2) - (2 \times 1.5 \times 1.8 \times \cos 112°)$ ✓

$a^2 = 7.51287$ ✓

$a = \sqrt{7.51287}$

$a = 2.74$ km ✓

Question 12 (2 marks)
What is the area of the above race course?

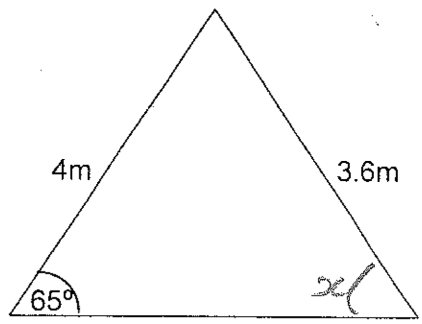
$A = \frac{1}{2}ab\sin C$ ✓

$= 0.5 \times 1.5 \times 1.8 \times \sin 112°$

$= 1.25\ \text{km}^2$ ✓

Question 13 (2 marks)
A step ladder (which folds near the middle) has the step side 4 m long making an angle of ~~70°~~ 65° with the level surface it is sitting on. The support side (back) is ~~3.8~~ 3.6 m long.

What angle does this side make with the ground?



$\frac{\sin x}{3.6} = \frac{\sin 65°}{4}$

$\cancel{4\times}\sin x = \frac{3.6 \times \sin 65}{4}$ ✓

$x = \sin^{-1}\left(\frac{3.6 \times \sin 65}{4}\right)$

$\therefore \theta$ or $x = 54.65 \approx \underline{\underline{54.7°}}$

Question 14 (7 marks: 2, 3, 2)

Eddie has won \$1 000 000.00 in the lottery!!
He asks the bank for an investment plan. The manager proposes 3 different plans.

- PLAN A earns 6.9% p.a. simple interest per year.
- PLAN B earns 5% interest per year, compounded annually.
- PLAN C earns 4.9% p.a. interest compounded quarterly.

a) How much does Eddy have at the end of 10 years for PLAN A?

$P \times R \times T = 1000000 \times 0.069 \times 10 \quad + (1\,000\,000)$

\$1, 690, 000

b) Eddy has heard that compound interest is better. If Eddy is to invest for 3 years, which is the better option PLAN B, or PLAN C?

Plan B

$A = P(1+i)^n$
$= 1000000(1.05)^3$
$= 1.15763 \times 10^6 = \$1\,157\,630$

(B is BEST OPTION)

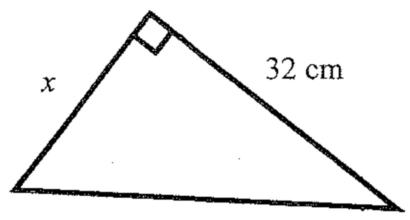
Plan C
$n = 4 \times 3 = 12$
$i = \frac{0.049}{4}$
$= 0.01225$

$A = 1000000(1.01225)^{12}$
$= \$1,157,320$

c) How much extra will he save?

$$\begin{array}{r} 1\,157\,630 \\ -\ 1\,157\,320 \\ \hline \$310 \end{array}$$

Question 15 (2 marks)
Find the value of the unknown in the triangle correct to the nearest mm.



By Pythagoras' Theorem

$x^2 = 43^2 - 32^2$

$x = \sqrt{(43^2 - 32^2)}$

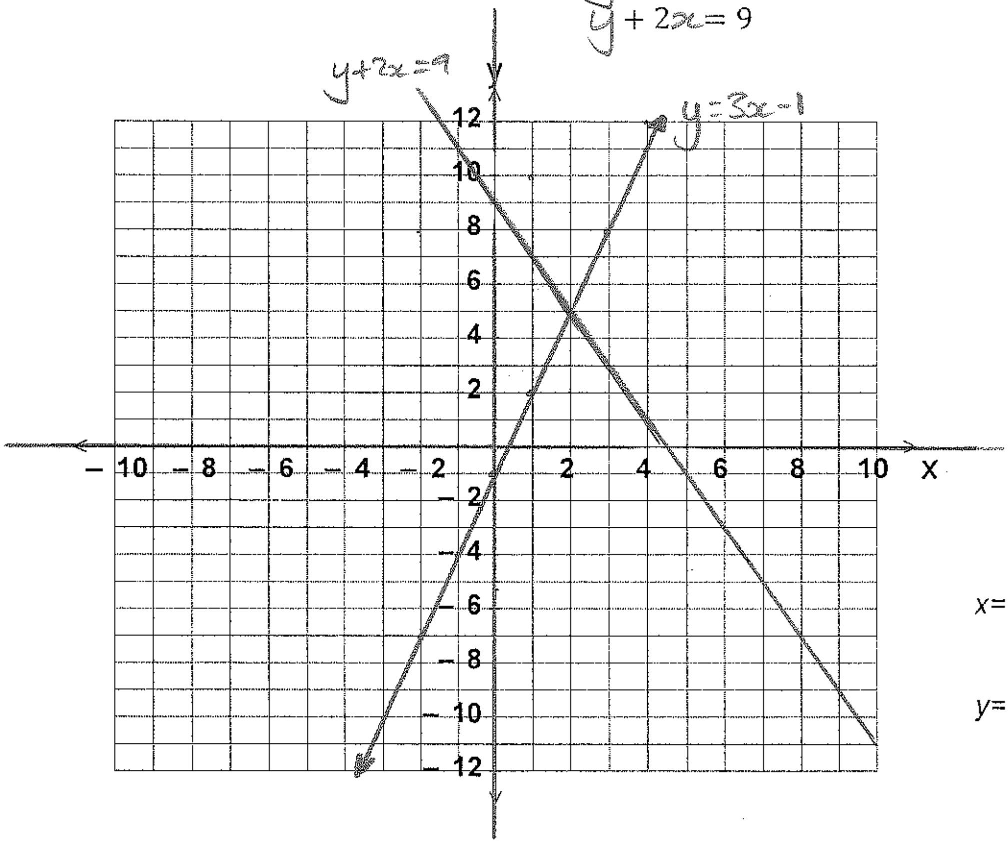
$= 28.7228$

≈ 28.7 cm to nearest mm

or 287mm

Question 16 (3 marks)
Solve the set of simultaneous equations by graphing.(Make the axes clearer by ruling them first)

$$y = 3x - 1$$
$$y + 2x = 9$$



1 for line
1 for line
1 for (2, 5)

$x = 2$

$y = 5$

Question 17 (5 marks: 2, 1, 2)

a) Find the distance between the points (-3, 5) and (6, -1), rounded to 2 decimal places

$\sqrt{(5 - -1)^2 + (-3 - 6)^2}$ ✓

$= \sqrt{36 + 81}$

$= \sqrt{117} = 10.8167$ ½

≈ 10.82 (-½ for rounding error)

b) Find the midpoint of the line joining (-2, 8) and (4, 12)

$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \left(\frac{-2+4}{2}, \frac{8+12}{2}\right) = (1, 10)$ ✓

c) Find the equation of the line that passes through the points (1, 5) and (5, 7)

$\frac{\Delta y}{\Delta x} = \frac{7-5}{5-1} = \frac{2}{4} = \frac{1}{2}$

$y = \frac{1}{2}x + c$

At (1, 5)

$5 = \frac{1}{2}(1) + c$

$c = 4\frac{1}{2}$

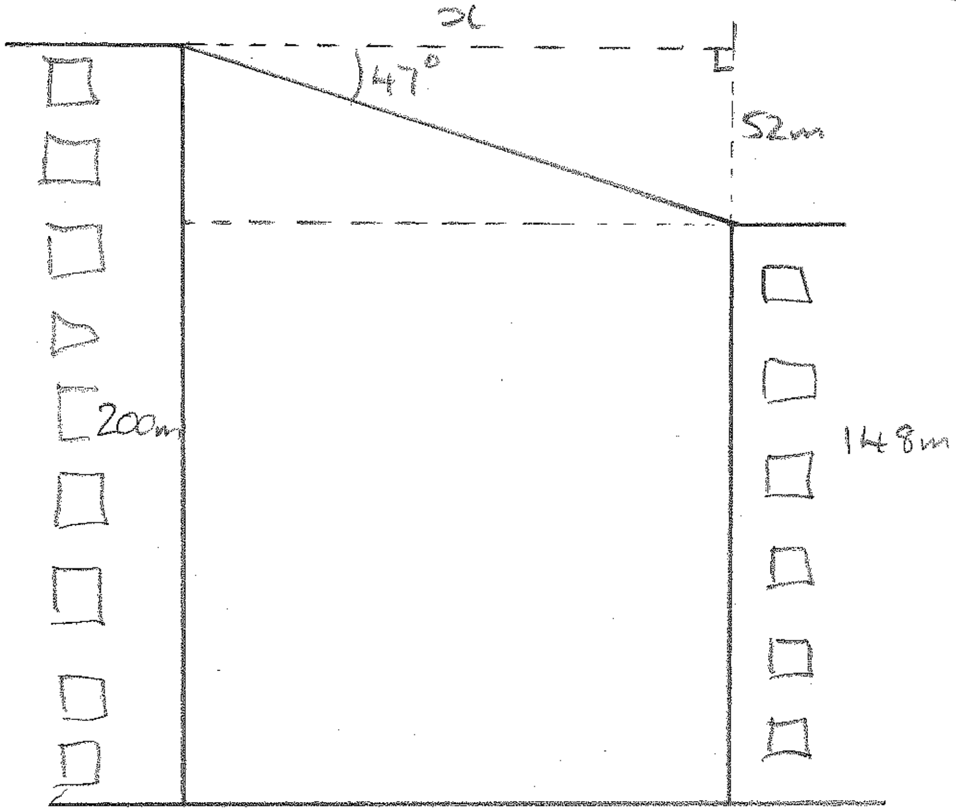
$\longrightarrow y = \frac{1}{2}x + 4.5$ ✓ ✓

or $2y = x + 9$ ✓✓

Question 18 (3 marks: 1,2)

Two buildings are directly opposite each other, either side of a street. One building is 200m tall and the other is 148m tall. The angle of depression from the top of the taller building to the top of the smaller building is 47°. How far apart are the two buildings? (Hint: Draw a diagram and solve the trigonometry)

$$\begin{array}{r} 200 \\ -148 \\ \hline 52 \end{array}$$



1 diagram

1 working

1 answer

$$\tan 47° = \frac{52}{x}$$

$$x = \frac{52}{\tan 47°}$$

$$= \underline{\underline{48.5m}}$$

End of Section 2

Spare Page